Assignment 1: Getting Started

In this assignment, you will implement convolutional kernels manually with NumPy, then verify each implementation with TensorFlow/Keras. The final task introduces forward propagation, L2 loss, gradient computation, and one SGD update step using TensorFlow. For Tasks, please see .ipynb file.

1 Setup

Before starting this assignment, you must prepare a Python environment in which you can run **NumPy**, **TensorFlow/Keras**, and supporting libraries such as **Matplotlib**.

For this course, you may use one of the following three options:

1. **Kaggle Notebooks** (**first priority, strongly recommended**)
2. **Google Colab** (**second priority, also recommended**)
3. **Local environment with GPU support** (**advanced option**)

For most students, **Kaggle Notebooks** is the preferred option because it is browser-based, easy to use, already contains many machine learning libraries, and can provide GPU access without difficult installation.

If one option does not work well for you, move to the next option.

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2 Recommended order of use

Use the following order:

1. **Kaggle Notebooks**
2. **Google Colab**
3. **Local GPU environment**

If you have trouble with the local environment, do **not** spend too much time debugging installation. Move to **Kaggle** or **Google Colab** and continue the assignment there.

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3 Required libraries

For this assignment, you will need the following Python libraries:

- `numpy`
- `tensorflow`
- `matplotlib`

- `pandas`
- `scikit-learn`
- `notebook`
- `jupyter`

In Kaggle and Colab, many of these are already installed.

If installation is required, you may install libraries individually using `pip`, or install all libraries from a `requirements.txt` file.

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4 Option 1: Kaggle Notebooks (First Priority)

Kaggle Notebooks allow you to write and run Python code in a browser without installing software on your own computer.

4.1 Create or sign in to a Kaggle account

1. Open your web browser.
2. Go to **Kaggle**.
3. Click **Sign In** if you already have an account.
4. If you do not have an account, click **Register** and create one.

4.2 Open the notebook interface

1. After signing in, look at the top navigation bar.
2. Click **Code**.
3. Click **Create New Notebook**.

A new notebook page will open.

4.3 Rename the notebook

1. At the top of the notebook page, click the notebook title.
2. Rename it to something meaningful, for example:

```
COMPX326_Assignment1.ipynb
```

Even if Kaggle does not show the file extension directly in the title, you should treat it as a notebook file (`.ipynb`).

4.4 Enable GPU in Kaggle

You should enable GPU before running TensorFlow tasks.

1. On the right side of the notebook page, find the **Settings** panel.
2. In the **Settings** panel, look for **Accelerator**.
3. Click the dropdown menu next to **Accelerator**.
4. Select **GPU**.
5. Wait while Kaggle prepares the notebook runtime.

4.5 Turn internet on or off if needed (optional)

Usually this assignment does not require internet access after setup, but you may check the setting.

1. In the right-side **Settings** panel, find **Internet**.
2. Turn it **on** only if you need to install missing libraries or download files.
3. Keep it **off** if not needed.

4.6 Upload notebook files or supporting files if needed

Import assignment notebook (IMPORTANT)

If you are provided with a notebook file (for example: assignment1.ipynb), you should import it into Kaggle instead of creating a new notebook.

Follow these steps:

1. Go to the Kaggle website.
2. Click on the Code tab.
3. Click Create New Notebook.
4. In the top menu, click File.
5. Click Import Notebook.
6. Select the .ipynb file from your computer.
7. Wait for the notebook to upload and open.

You will now see the full assignment notebook inside Kaggle and can start working on it.

Add additional input files (datasets, resources)

If you are provided with extra files such as:

- dataset files
- images
- additional resources
- helper files
- requirements.txt

you can add them to your notebook as input files.

These files are used as additional resources that your notebook can read.

Steps to add input files

1. On the right side of the notebook, locate the Add Input section.
2. Click Add Input.
3. Choose one of the following options:
   - Upload (to upload files from your computer), or
   - Select an existing dataset from Kaggle.
4. Select the required files.
5. Wait until the files are attached to the notebook.

Important note about input files

- Files added using Add Input are read-only.
- You can open and use them in your notebook.
- You cannot directly modify these files.
- If you need to modify a file, you must first copy it to your working directory.

4.7 Test whether TensorFlow is available

In the first code cell, type the following:

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt

print("TensorFlow version:", tf.__version__)
print("NumPy version:", np.__version__)
print("GPU devices:", tf.config.list_physical_devices('GPU'))
```

To run the cell:

1. Click inside the cell.
2. Press **Shift + Enter**.

If the environment is working, you should see:

- a TensorFlow version number
- a NumPy version number
- a list of GPU devices, or an empty list if no GPU is assigned

4.8 Install libraries individually with pip if needed

If a library is missing, create a new cell and run:

```
!pip install numpy tensorflow matplotlib pandas scikit-learn notebook jupyter
```

Then test again.

4.9 Install libraries from a requirements.txt file if provided

If you have a `requirements.txt` file:

1. Upload the file to Kaggle.
2. In a code cell, run:

```
!pip install -r requirements.txt
```

4.10 Recommended Kaggle verification cell

Run the following to confirm everything is ready:

```
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

print("NumPy version:", np.__version__)
print("TensorFlow version:", tf.__version__)
print("GPU devices:", tf.config.list_physical_devices('GPU'))

x = np.array([1, 2, 3])
print("Test array:", x)

print("Kaggle setup successful.")
```

If this runs without errors, your Kaggle setup is ready.

4.11 Save your work in Kaggle

Kaggle notebooks save automatically, but you should still check.

To save a version:

1. Click **Save Version** near the top-right of the notebook page.
2. Wait until Kaggle finishes saving.

To download a notebook copy:

1. Click the **File** or **More options** menu if available.
2. Choose **Download notebook** or export the notebook when needed.

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5 Option 2: Google Colab (Second Priority)

Google Colab is also a browser-based notebook environment and is a very good backup if Kaggle is not convenient.

5.1 Open Google Colab

1. Open your web browser.
2. Go to **Google Colab**.
3. Sign in with your Google account.

5.2 Create a new notebook

1. In the top menu, click **File**.
2. Click **New notebook**.

A new notebook will open in a new tab.

5.3 Rename the notebook

1. At the top-left, click the notebook title, which may look like `Untitled0.ipynb`.
2. Rename it to:

```
COMPX326_Assignment1.ipynb
```

5.4 Enable GPU in Google Colab

1. In the top menu, click **Runtime**.
2. Click **Change runtime type**.
3. In the dialog box, find **Hardware accelerator**.
4. Click the dropdown menu.
5. Select **GPU**.
6. Click **Save**.

5.5 Upload files if needed

If you are provided with files such as:

- starter notebook
- datasets
- images
- `requirements.txt`

then do the following:

1. On the left side, click the **folder icon**.
2. In the file panel, click the **upload icon**.
3. Select the files from your computer.
4. Wait for the upload to finish.

5.6 Verify the Colab environment

In the first code cell, run:

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt

print("TensorFlow version:", tf.__version__)
print("NumPy version:", np.__version__)
print("GPU devices:", tf.config.list_physical_devices('GPU'))
```

5.7 Install libraries individually with pip if needed

```
!pip install numpy tensorflow matplotlib pandas scikit-learn notebook jupyter
```

5.8 Install from requirements.txt if provided

```
!pip install -r requirements.txt
```

5.9 Recommended Colab verification cell

```
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

print("NumPy version:", np.__version__)
print("TensorFlow version:", tf.__version__)
print("GPU devices:", tf.config.list_physical_devices('GPU'))

x = np.array([1, 2, 3])
print("Test array:", x)

print("Google Colab setup successful.")
```

5.10 Save and download notebook in Colab

To save:

- Colab usually autosaves to Google Drive

To download:

1. Click **File**
2. Click **Download**
3. Choose **Download .ipynb**

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6 Option 3: Local environment with GPU support (Advanced)

This option is for students who want to work on their own computer and are comfortable installing software.

For this course, a local environment should be a **GPU-enabled environment** if possible.

If you do not have a compatible GPU or the setup becomes difficult, use **Kaggle** or **Google Colab** instead.

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7 Local system requirements

To use TensorFlow with GPU locally, your computer should have:

- an **NVIDIA GPU**
- installed **NVIDIA drivers**
- installed **CUDA Toolkit**
- installed **cuDNN**
- **Python 3**
- a notebook environment such as **Jupyter Notebook** or **VS Code with Jupyter support**

If your computer does not have a supported NVIDIA GPU, use **Kaggle Notebooks**.

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8 Check whether your computer has an NVIDIA GPU

On Windows

1. Right-click the **Start** button.
2. Click **Task Manager**.
3. Click the **Performance** tab.
4. Look for **GPU** on the left side.
5. Check whether it shows an NVIDIA GPU.

Another method:

1. Right-click the **Start** button.
2. Click **Device Manager**.
3. Expand **Display adapters**.

4. Check whether an NVIDIA graphics card is listed.

On Linux

Open terminal and run:

```
nvidia-smi
```

If GPU information appears, your system likely has NVIDIA GPU support.

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9 Create your COMPX326 folder

Create a folder for this course on your computer.

For example:

```
COMPX326/
```

A suggested notebook-based structure is:

```
COMPX326/
├── requirements.txt
├── assignment1_tensorflow.ipynb
├── assignment1_tensorflow_solution.ipynb
└── data/
```

If you use OneDrive, Google Drive, or another cloud backup system, you may store this folder there as well.

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10 Install Python

10.1 Download Python

1. Open your browser.
2. Go to the official Python website.
3. Download a recent version of **Python 3**.

10.2 Install Python on Windows

1. Open the installer.
2. Before clicking install, make sure to tick:

```
Add Python to PATH
```

3. Click **Install Now**.
4. Wait until installation is complete.

10.3 Verify Python installation

Open **Command Prompt** or **Terminal** and run:

```
python --version
```

If that does not work, try:

```
python3 --version
```

You should see a Python 3 version number.

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11 Install Jupyter Notebook or use VS Code with notebooks

You will work in **`.ipynb` notebook format**, not in `.py` scripts.

Option A: Jupyter Notebook

Install Jupyter with pip:

```
pip install notebook jupyter
```

Then later you can start Jupyter by running:

```
jupyter notebook
```

Option B: VS Code with Jupyter support

1. Open your browser.
2. Go to the official Visual Studio Code website.
3. Download and install VS Code.
4. Open VS Code.
5. Click the **Extensions** icon on the left.
6. Search for **Python** and click **Install**.
7. Search for **Jupyter** and click **Install**.

This will allow you to open and run `.ipynb` notebooks inside VS Code.

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12 Install NVIDIA drivers

12.1 Download and install driver

1. Open your browser.
2. Go to the official NVIDIA driver download page.
3. Select your GPU model.
4. Download the correct driver.
5. Run the installer.
6. Restart your computer if required.

12.2 Verify the driver

Open terminal or command prompt and run:

```
nvidia-smi
```

If successful, you should see:

- GPU model
- driver version
- CUDA-related information

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13 Install CUDA Toolkit

TensorFlow GPU support requires compatible CUDA and cuDNN versions.

13.1 Download CUDA Toolkit

1. Open your browser.
2. Go to the official NVIDIA CUDA Toolkit page.
3. Download a version compatible with your TensorFlow version.
4. Run the installer.
5. Complete the installation.

13.2 Verify CUDA

Open terminal and run:

```
nvcc --version
```

If successful, CUDA version information will be displayed.

14 Install cuDNN

cuDNN is required for GPU acceleration in deep learning.

14.1 Download cuDNN

1. Open your browser.
2. Go to the official NVIDIA cuDNN page.
3. Sign in if required.
4. Download a cuDNN version compatible with your CUDA installation.

14.2 Install cuDNN

Follow NVIDIA's official instructions for your operating system.

Usually this involves:

- extracting the downloaded files
- copying cuDNN files into the CUDA installation directories

Make sure CUDA and cuDNN versions are compatible.

15 Create and activate a virtual environment

A virtual environment helps keep this course separate from other Python projects.

15.1 Open terminal in your COMPX326 folder

Navigate to your `COMPX326` folder and open terminal there.

15.2 Create the environment

On Windows

```
python -m venv venv
```

On macOS/Linux

```
python3 -m venv venv
```

15.3 Activate the environment

On Windows

```
venv\Scripts\activate
```

On macOS/Linux

```
source venv/bin/activate
```

After activation, your terminal should show that the `venv` environment is active.

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16 Install the required libraries locally

You may install libraries in one of two ways.

16.1 Install specific libraries with pip

Run:

```
pip install numpy tensorflow matplotlib pandas scikit-learn notebook jupyter
```

16.2 Install from requirements.txt

If your lecturer provides a `requirements.txt` file, put it inside your `COMPX326` folder and run:

```
pip install -r requirements.txt
```

A sample `requirements.txt` file may contain:

```
numpy
tensorflow
matplotlib
pandas
scikit-learn
notebook
jupyter
```

---

17 Create or open the notebook file locally

You should work in a notebook file such as:

```
assignment1_tensorflow.ipynb
```

If using Jupyter Notebook

1. Open terminal in the `COMPX326` folder.
2. Run:

```
jupyter notebook
```

3. A browser page will open.
4. Click **New**.
5. Click **Python 3**.
6. Rename the notebook to:

```
assignment1_tensorflow.ipynb
```

If using VS Code

1. Open VS Code.
2. Click **File**.
3. Click **Open Folder...**
4. Select `COMPX326`.
5. In the left panel, click **New File**.
6. Name it:

```
assignment1_tensorflow.ipynb
```

7. Open the notebook.
8. Choose the Python interpreter connected to your virtual environment.

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18 Verify local TensorFlow and GPU setup

In the first notebook cell, run:

```
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

print("NumPy version:", np.__version__)
print("TensorFlow version:", tf.__version__)
print("GPU devices:", tf.config.list_physical_devices('GPU'))
```

If everything is correct, TensorFlow should show one or more GPU devices.

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19 Recommended full verification cell for all environments

No matter which platform you use, run the following cell before starting the assignment:

```
import numpy as np
import tensorflow as tf
import matplotlib.pyplot as plt

print("NumPy version:", np.__version__)
print("TensorFlow version:", tf.__version__)
print("GPU devices:", tf.config.list_physical_devices('GPU'))

x = np.array([1, 2, 3])
print("Test array:", x)

a = tf.constant([[1.0, 2.0], [3.0, 4.0]])
b = tf.constant([[5.0, 6.0], [7.0, 8.0]])
print("TensorFlow test result:")
print(tf.matmul(a, b))

print("Environment setup successful.")
```

If this runs without errors, your environment is ready.

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20 Saving your work

On Kaggle

- Kaggle usually autosaves
- Click **Save Version** to create a saved version

On Google Colab

- Colab usually autosaves to Google Drive
- To download, click **File → Download → Download .ipynb**

On local Jupyter or VS Code

- Save the notebook regularly using **Ctrl + S**
- Keep backup copies if needed

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21 Final recommendation for students

For this assignment, use the following order:

1. **Kaggle Notebooks**
2. **Google Colab**
3. **Local GPU environment**

If you are unsure which one to choose, use **Kaggle** first.

If local installation becomes difficult, do not waste too much time on setup. Switch to **Kaggle** or **Google Colab** and continue the assignment there.